Campus Store — Quick Summary & Function Guide

Live Web Link: https://naveenjaay55-cpu.github.io/OJT/ | College OJT Project

1. Two Roles & Permissions

Role	What They Can Do	Restrictions (Enforced in Code)
Cashier (C001)	Select items, check stock, complete checkout, print receipts.	Cannot add items, edit prices, or change stock.
Store Staff (S001)	Add new products, update prices, restock quantities, set alerts.	Cannot complete checkouts at POS terminal.

2. Key Features (In Simple English)

- **Product Catalog (FR-01):** Shows all products with ID, name, price, and current stock.
- **Live Checkout (FR-04 to FR-07):** Select item, enter quantity; total updates live (Price × Qty).
- **Automatic Stock Decrement (FR-08, FR-09):** Stock decreases immediately after purchase.
- **Prevent Overselling (FR-10):** If quantity > stock, purchase is rejected and stock stays unchanged.
- **Low-Stock Alert (FR-11):** Yellow warning badge appears when stock <= alert threshold.
- **Local Storage (FR-12):** Auto-saves all products and orders in browser localStorage.
- **Sales Summary & History (FR-13, FR-14):** Shows total revenue, order count, and past receipts.

3. Which Function Does Which Work?

File	Function Name	What It Does in Simple Words
services.js	**loadData() / saveData()**	Loads and saves products and orders in browser localStorage.
services.js	**addProduct(product)**	Adds a new product to inventory (Staff only).
services.js	**updateProduct(id, updates)**	Edits product name, price, or stock (Staff only).
services.js	**processCheckout(id, qty)**	Checks stock, deducts inventory, records sale (Cashier only).
services.js	**getLowStockProducts()**	Finds all products that have low stock.
services.js	**getSalesSummary()**	Calculates total revenue and completed order count.
services.js	**setCurrentRole(role)**	Switches role between Cashier and Store Staff.
app.js	**renderPOSProducts()**	Draws the visual product cards on the screen.
app.js	**updateLiveCalculation()**	Calculates Total Amount (Price × Quantity) live.
app.js	**handleProcessCheckout()**	Runs checkout when button clicked; shows receipt on success.
app.js	**handleTestOversell()**	1-click demo button that tests stock limit rejection.
app.js	**renderInventoryTable()**	Draws inventory table with Edit and Delete buttons.
app.js	**syncRoleUI()**	Locks/unlocks buttons based on whether you are Cashier or Staff.
replay_checkout.py	**run_replay()**	Python script testing sequential purchases and asserting stock >= 0.